Reinforce

Kanish H R, University of Edinburgh, s2791283@ed.ac.uk

Abstract

In engineering and science courses, lecturers and tutors usually learn what a class finds difficult late, from lectures or after hand-ins, while students are left with a standard set of resources. We propose an adaptive mastery-learning web app built on the course's own textbook. A language model drafts a concept graph from the book, and the professor reviews and approves it. Students answer auto-graded questions tagged to concepts. Bayesian knowledge tracing updates a mastery estimate for each concept after every answer, the next questions target the weakest concepts, and a concept unlocks only when a conservative lower bound on that estimate is high enough. All of this sits inside a chatbot harness, and the professor sees which concepts the class finds hard and which wrong answers are common. The intended impact is lower setup cost for the professor and earlier, concept-level visibility of difficulty.

1. Introduction / Problem Statement

Lecturers usually find out what a class struggles with late: through questions in lectures, or in tutorials/workshops, when little time is left to respond. Furthermore, the growing use of AI has also made this unreliable to an extent, when coursework starts affecting grades. Students meanwhile receive the same problem sheet whatever they already know, so a weak topic can go unnoticed until it blocks later material, in a subject where each chapter builds on the last. Thus, it becomes a key problem identifying whether the student is actively learning from a formative perspective.

Who is affected. Students (focusing on the quantitative courses for simplicity of the proposal), and the lecturers and tutors who run their tutorials.

Why it matters. With the significant cognitive offloading seen in learning, there arises a clear need for increased focus on mastery-based practice. Here, competency means understanding a concept and mastery means being able to use it comfortably. Mastery-based practice, where students move on once they show they know a topic, produced gains of about 0.18–0.22 standard deviations in randomised school-maths trials [1, 2]. A review found step-based computer tutoring reached $d \approx 0.76$ against no tutoring [3]. These are results for other systems and subjects, not for ours. What stops most courses using such practice is set-up: someone must split the course into concepts, order them and tag every question. Our bet is that a language model can draft that from the textbook the professor already uses, leaving the professor to review it. Quantitative subjects like Physics or a branch of Engineering suit the approach because numerical answers with units can be marked automatically (this also avoids having to handle the issue of luck playing a factor in getting the right answer when using the BKT). This map-based unlocking is a mild form of gamification, which one study of 133 undergraduates linked to higher motivation [10].

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2. Proposed Solution

a) The process it supports. A language model drafts and the professor decides. There are six stages.

1. **Concept graph.** The professor picks a main textbook. The system extracts its sections and cross-references, and a language model proposes prerequisite links between concepts, each shown with its evidence. The professor now has the provision to accept, reject, or retype certain prerequisites, then approves the graph. Nothing restricts students before approval, and manual edits survive re-runs.

2. **Question bank.** Textbook exercises are transcribed, tagged to at most three concepts, and given a difficulty estimate and a short ladder of pre-written hints. Low-confidence items go to a review queue.

3. **Targeted practice.** Students work through concepts on a map. Only the first attempt at a question counts, and hints used are logged. Each answer updates the student's mastery estimates for every concept the question is tagged with. The next questions target the weakest estimates, mixed with stronger ones so practice is not a run of failures. A checkpoint closes each concept (a chapter or section covering several subtopics): a short, supervised test in a tutorial, as seen in Y3+ courses in Engineering, of the student's mastery of that concept. It avoids passive learning and the cognitive offloading that the individual might get from using an unrestricted chatbot.

4. **Conservative unlocking.** A concept counts as mastered only when the low end of an uncertainty range passes a threshold and at least three different questions have been tried first time. A locked concept shows why, and a professor can override any lock. The checkpoint at the end of a concept (stage 3) confirms the estimate under supervision. Because concepts sit on a prerequisite map and open only as mastery is shown, progress works as a mild form of gamification: clearing one node opens the next.

5. **Harness.** This system is wrapped in a harness, with a chatbot that releases the pre-written hints one step at a time, guiding the student towards the answer if they are stuck, and keeps track of the number of hints used by the student.

6. **Professor view.** For each concept, the class error rate and the most common wrong answer, so tutorial time goes where students struggle.

b) The technology. Language models draft the prerequisite links, tags, difficulty estimates and hints, behind one provider interface with caching and a spending cap. Mastery is tracked with Bayesian knowledge tracing (BKT) [9], which keeps a probability for each concept that the student knows it. Each question's difficulty starts as a language-model estimate. As responses accumulate, it could be re-estimated with Item Response Theory (IRT), which fits question difficulty from many students' answers, so that a hard question does not over-blame a concept.

A question tagged with several concepts is treated as needing all of them, so a wrong answer moves most of the blame to the concept already estimated as weaker. Next questions are chosen from these estimates directly, not from separate reinforcement weights, which would be a second

number to drift out of step with the model. In a calculation of our own with illustrative parameters, three correct answers already push the point estimate past 0.95, so the lower bound and the three-question floor do the real gating. Hints are checked for leaked answers, and numeric answers are marked with a tolerance and units.

Current status. Built, on synthetic data: the student concept map, the professor graph-review screen and the backend.

c) Links

- Code: https://github.com/potatoesandfrenchfries/teachathon-26-demo

3. Implications & Ethical Considerations

Benefits. Less set-up work for the professor (an expectation, not a measurement), earlier concept-level visibility of difficulty, practice aimed at each student's weak topics, and a model simple enough to inspect. The AI proposes and a human decides: the professor approves the graph and tags, every proposed link shows its evidence, and a locked concept shows why. The tool sits beside lectures and tutorials, and we propose completion and not correctness, as the only credit given to students.

This can also fundamentally modify how tutorials/workshops are being offered, as we have seen the recent shift in students having to do questions in front of tutors to display whether they understand the concepts for Y3+ courses at the School of Engineering. This serves well for the purposes of checking competency, while this application checks each student's mastery of every concept beforehand, through formative practice.

Risks and mitigations

- **Unfair or wrong locks.** Mastery thresholds can disadvantage students whose first attempts are noisy for reasons other than knowledge, such as time, language or device. We have not tested this. Early on, question difficulty is only a model's guess, so a hard item can over-blame a concept. About a third of students may never reach mastery by repeating items [4], so we use soft gating with overrides, plan to report who is blocked where, and plan a "needs a person" flag.
- **Wrong tags.** A wrongly tagged question biases that question's parameters for every student [8], and evidence on language-model tagging is early. We use a closed list of concepts, a cap of three tags, a professor review queue and, as planned work, a separate algebra and units tag so that arithmetic slips do not blame the concept/skill.
- **Model errors in hints.** In one study, 32% of generated help failed a quality check before filtering [6]. We check every hint ladder for leaked answers, and a person skims each one.
- **Over-reliance on AI and academic integrity.** High-school students with unrestricted GPT-4 access scored 17% worse on a later exam [5]. Our tool is for practice and diagnosis with scripted hints, not open chat. Unsupervised home answers may still reflect chatbot use, so supervised checkpoints are the check on validity. Reported gains for AI tutors are contested: a re-analysis of one physics trial [7] found a much smaller advantage among completers. It is always useful to separate competency from mastery of a concept in formative testing.

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- **A misleading dashboard.** Voluntary use biases the board towards engaged students, and a failed numerical problem often reflects algebra or units, not the tagged concept. This selection bias could be addressed by offering an incentive to use the platform, with the checkpoints being used for the tutorials, encouraging the student to actually practice the other sections.
- **Privacy, licensing and access.** No names or emails are collected, students can delete their data, and individual views are opt-in. A real deployment would need a data protection impact assessment, and students should be told what is kept, for how long and who can see it. The main book is assumed to be openly licensed. Other books are internal-only and never exported. The model provider sits between us and the model, and we have not checked its data terms for licensed pages. A web app also disadvantages students without a reliable device or connection, which we have not tested.

Unintended consequences. Students may click through hints, so we log time between hint levels and flag the pattern for the professor without accusing anyone. Professors may over-trust a dashboard built on a small, self-selected cohort. Nothing here shows that acting on the dashboard improves learning. That needs a pilot.

References

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